Paleomagnetism of Anorthosite in and around the Charlevoix Cryptoexplosion Structure, Quebec

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Grenville anorthosite more than 20 km away from the center of the Charlevoix structure appears unaffected by the deformation, and the mean remanent vector from five sites, after demagnetization, is $D = 128$, $I = 57$, $k = 49$, $\alpha_{95} = 11$; pole = 7° N Lat., 32° W Long., $\delta p = 12°$, $\delta m = 16°$. Progressive rotation of the NRM at three sites between 15 and 20 km of the structure center is compatible with central doming; the vectors from three sites within 15 km are anomalous, possibly reflecting modification (reversal?) of the primary vector, plus considerable tectonic rotation. The NRM at one site in 'impactite' (?), radiometrically dated at 350 m.y., is $n = 3$, $D = 337$, $I = 40$, $k = 22$, $\alpha_{95} = 27$.

L'anorthosite de Grenville à plus de 20 km du centre de la structure de Charlevois n'apparaît pas affectée par la déformation, et le vecteur rémanent moyen de cinq sîtes est, après démagnétisation, $D = 128$, $I = 57$, $k = 49$, $\alpha_{95} = 11$; le pôle ayant pour latitude 70°N et 32°W pour longitude, $\delta p = 12°$, $\delta m = 16°$. Il est trois sîtes entre 15 et 20 km du centre de la structure où la rotation progressive du NRM est compatible avec le bombement central; les vecteurs de trois sîtes dans un rayon de 15 km sont irréguliers, réfléchissant peut-être une modification (renversement?) du vecteur primaire, en plus de la rotation tectonique considérable. Le NRM d'un sîte pris dans une "impactite" (?), datée radiométriquement à 350 millions d'années, est $n = 3$, $D = 337$, $I = 40$, $k = 22$, $\alpha_{95} = 27$.

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Introduction

The Charlevoix cryptoexplosion structure (Rondot 1966, 1968, 1971, 1972; Robertson 1968) centered around Mont des Eboulement (Fig. 1) is located on the north shore of the St. Lawrence River, 110 km northeast (downstream) from Quebec City. Prior to the recognition of the cryptoexplosion structure by Rondot and Robertson, the general area was best known for the titanium deposits in anorthosite at St. Urbain (Mawdsley 1927; Rose 1962, 1972). The basement terrain consists of charnockitic and amphibolitic gneisses of Grenville age, with domical anorthosite masses of varying dimension. Paleozoic rocks (middle and upper Ordovician limestones and clastics) are found only along the St. Lawrence River, and in the ring syncline surrounding Mont des Eboulements, the central uplift of the structure (see Fig. 1).

As the lower Paleozoic rocks in and around the structure are considerably deformed and shock-metamorphosed, the structure is known to have formed post early upper Ordovician (Rondot 1972). The next sedimentary deposits however, are Pleistocene in age, and hence the time of formation of the structure cannot be accurately constrained on geologic grounds. Potassium–argon radiometric ages on samples and minerals from an outcrop of fine grained heterogeneous igneous rock, thought to be an 'impactite' (Rondot 1971), average 350 m.y., which is the best current estimate of the time of formation.

Sampling and Measurement

In addition to the impactite, oriented samples (principally of anorthosite which, as a rock type, has proved to have excellent remanent magnetic properties) were collected for paleomagnetic study from outcrops ranging from Mont des Eboulements, the center of the structure, to in excess of 40 km from this point. The location of all sites are shown in Fig. 1. Samples, some drilled in the field, some hand-samples, were oriented either by suncompass, or magnetic compass plus sighting on prominent topographic features; details as to the petrology are given in Table 1.

The Natural Remanent Magnetism (NRM) of all samples was measured on a Princeton

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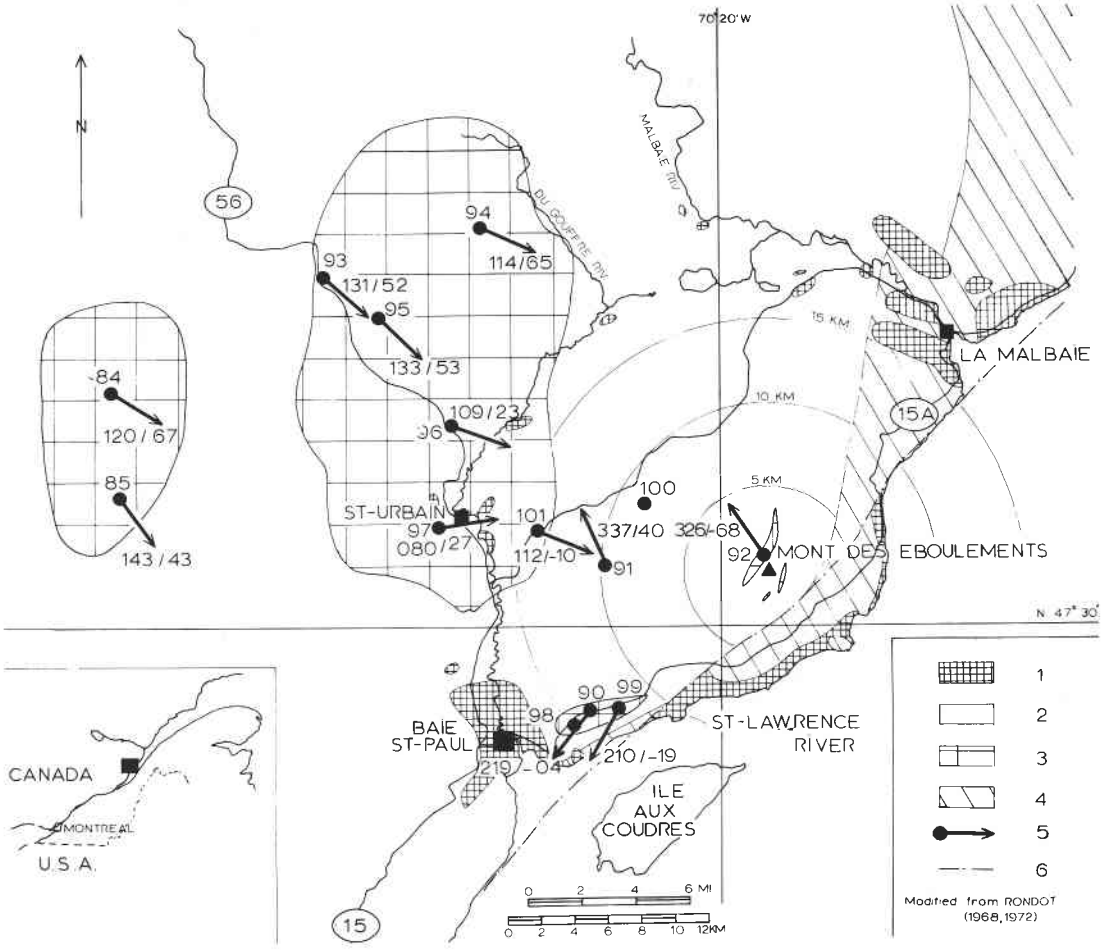


FIG. 1. Location, generalized geology, sampling sites, and paleomagnetic vectors in and around the Charlevoix cryptoexplosion structure. Mont des Eboulements is at the center of the structure. (1) Paleozoic sediments, (2) charnockitic gneisses, (3) anorthosite, (4) granitic and paragneissic sequence, (5) sampling site (solid circle) and remanent vector (arrow), numbering, as 337/40, gives declination and inclination, north-seeking pole down is positive, (6) trace of the principal step of the normal step-fault along the St. Lawrence north shore.

Applied Research S.M. 1 spinner magnetometer. Pilot specimens from each site were systematically demagnetized in alternating fields (AF) up to 500 oe, the optimum demagnetizing field for treating the remaining samples being inferred from these reconnaissance measurements.

Results

The magnetic results obtained for all 14 sites are shown in Table 1. The natural remanence of the impactite (site 91) and the highly shocked anorthosite from the crest of Mont des Eboulement (site 92) were weak, and highly inconsistent, but after intensive demagnetization to 500 oe, repeated measurement, and rejection of one sample from each site, a consistent vector was obtained. Only the charnockite (site 100) and one anorthosite (site 98) failed to respond to such treatment.

The NRM of most of the other ten sites, while varying in intensity from 1×10^{-3} to 5×10^{-6} emu/cc, was remarkably resistant to AF demagnetization, in most cases being almost completely unaffected in direction and intensity by demagnetizing fields up to 500 oe (Fig. 2).

The exclusive, or overwhelmingly predomi-

TABLE 1. Results of paleomagnetic measurements

Site	Rock type	Samples	NRM D	NRM I	NRM k	NRM α_{95}	Site	H(oe)	AC demagnetized D	AC demagnetized I	AC demagnetized k	AC demagnetized α_{95}
84	Anorthosite	4	119.5	69.3	239.4	5.0	84	300	120.2	67.3	260.7	4.7
85	Anorthosite	4	137.9	51.4	26.8	18.1	85	300	142.8	43.4	459.0	4.3
90	Anorthosite	4	218.7	−1.8	41.1	14.5	90	300	218.8	−4.3	43.8	14.0
91	Impactite	4	scattered				91	500	336.9	39.6	22.3	26.7
92	Shocked anorthosite	4	scattered				92	500	325.8	−68.2	13.2	35.3
93	Anorthosite	4	128.9	56.3	58.9	12.1	93	300	131.4	52.2	80.1	10.3
94	Anorthosite	4	113.3	65.5	23.5	19.4	94	300	113.8	65.0	23.0	19.6
95	Anorthosite	5	50.1	5.9	1.3	90.0	95	500	132.6	52.6	15.9	19.8
96	Anorthosite	4	109.0	25.1	61.5	11.8	96	300	109.4	23.1	61.1	11.8
97	Anorthosite	4	79.7	31.1	17.7	22.5	97	300	80.2	27.0	16.9	23.0
98	Anorthosite	5	288.7	64.8	3.5	48.1	98	2/300	288.9	51.5	2.4	64.7
99	Anorthosite	4	218.1	0.8	6.0	41.0	99	300	209.7	−18.8	31.4	16.7
100	Charnockite, with shatter cones	4	318.6	82.4	8.4	33.7	100	100	11.7	56.3	1.6	90.0
101	Anorthosite	4	111.4	−10.5	100.5	9.2	101	300	112.2	−10.4	109.2	8.8

Locality mean vector after demagnetization

Sites	N	D	I	k	α_{95}	Pole Lat	Pole Long	α_{lat}	α_{long}
84, 85, 93, 94, 95	5	127.9	57.0	48.8	11.1	7°N	32°W	12	16

D = declination.
I = inclination.
k = Fisher's statistical parameter.
α_{95} = ½ angle of cone of 95% confidence around the vector mean.
N = number of units (samples or sites) used in the averaging.
H(oe) = intensity of AC demagnetizing field.

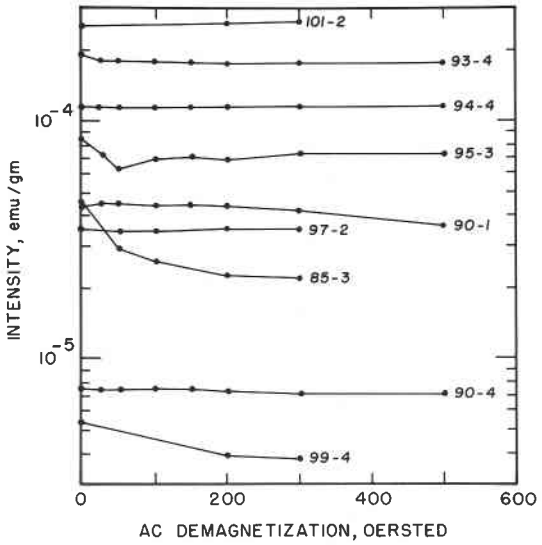


FIG. 2. Variation in specific remanent intensity with increasing AC demagnetization in anorthosite samples from Charlevoix. Specimens 90-4 and 99-4 are from the shocked anorthosite sites with anomalous directions; although weaker than average anorthosite, they are of comparable relative stability.

nant accessory Fe–Ti oxide mineral in these anorthosite rocks is an exsolved intergrowth of titanhematite and ferrian ilmenite; hematite is usually the major component of the intergrowth. The intensity of NRM correlates positively with the modal abundance of this phase. The ilmenite in all samples shows variable, but conspicuous alteration to a cryptocrystalline aggregate with much internal reflection, which is presumably metailmenite, composed of hematite and rutile or anatase. By using magnetic colloid (Elmore 1938) sparse, and very thin strongly magnetic; lamellae (or rods?) of what is presumed to be magnetite ($<1\mu$ to 10μ) were detected parallel to (0001) of some rhombohedral host grains, closely associated with, or apparently replacing hematite lamellae. The temperature dependence of high field magnetic properties (J_s/T) gave a Curie point of around 580 °C, but whether this was due to pure Fe_3O_4 in the lamellae, or lamellae of hematite with about 12% ilmenite still in solid solution, has not yet been positively determined. As such magnetite rod-lamellae are likely to be pseudosingle domain (Stacey and Banerjee (in press)) and titanhematite typically has high coercivity, either of these could equally well account for the exceptional stability of the NRM in the anorthositic rocks around St. Urbain.

Discussion of Vector Directions

The average declination and inclination (+ = N seeking pole down) at each site (Table 1) are shown in Fig.1. Greatest between site consistency is recorded by the five sites located more than 20 km from the center of the cryptoexplosion structure. The average vector from these five sites, which is assumed to be the primary RM, gives a VGP (Lat. 7° N, Long. 32° W), which is generally similar to that reported from other Grenville localities (Irving *et al.* 1972). The fact that these remanent vectors seem to point directly towards the Charlevoix structure (see Fig. 1) is considered to be fortuitous.

Sites 96 and 97, approximately 20 km from Mont des Eboulements, have generally similar RM declinations but the positive inclination is markedly shallower (average 25°); at site 101, 15 km from the center, the remanent vector points 10°*above* the horizontal.

This change in the remanent vector progressing inwards along a general NW/SE radius toward Mont des Eboulements, suggests a tectonic rotation associated with the central uplift and the peripheral graben.

Sites 90 and 99, just east of Baie St. Paul and 12 to 14 km from the structural center, are similar to one another, but markedly different both in declination and inclination from the primary remanent direction. The vector at site 92, (highly shocked anorthosite from the crest of Mont des Eboulement, $D = 326$, $I = -68$) is almost antiparallel to the primary regional vector ($D = 128$, $I = +57$).

The significance of these completely anomalous vectors (sites 90, 92, 99) is not clear. Possible explanations considered are: (1) remagnetization at time of formation of the structure, either (a) parallel to the ambient field, or (b) geometrically related to the direction of shock propagation; (2) tectonic rotation of the original vector.

The ambient field direction at Charlevoix in lower Paleozoic time judging from the Silurian pole position (Table 2) must have been generally due north/south with shallow inclinations, negating the possibility of simple com-

TABLE 2. Mid-Paleozoic pole positions for North America (Irving and Park 1972), contrasted with that obtained from the contemporary geomagnetic field at Charlevoix, and from the NRM in the Charlevoix impactite (site 91)

	Pole position	
	Latitude	Longitude
Silurian	32N	102E
Devonian	29N	110E
Carboniferous	36N	127E
Current Local V.G.P.	72N	101W
Site 91	59N	154E

plete remagnetization without at least some subsequent tectonic rotation. Although the possibility of partial remagnetization of the original Precambrian direction in Silurian times cannot be excluded, there is no evidence to suggest the presence of two separate vector components in the demagnetization data (Fig. 2).

The remanent vector at sites 90 and 99 is almost directly radially away from the structure center; the vector at site 92 may be regarded similarly if the shock locus had been below Mont des Eboulements. No evidence of remagnetization geometrically related to shock propagation was found, however, by Hargraves and Perkins (1968) in their investigation of RM in diverse, naturally and experimentally shocked rocks, and there is little else to favor this explanation at Charlevoix.

The progressive tectonic rotation of the primary RM vector apparent at sites 96, 97, and 101 (Fig. 1) encourages consideration of the tectonic explanation for the highly anomalous vectors at sites 90/99 and 92, but no simple updoming or overturning model is adequate. Paleozoic sediments in the ring-syncline east of Baie St. Paul, and just west of Sites 90/99, have been upturned to slightly overturned to the *west* or *southwest*. But if the anorthosite nearby is considered to be similarly upturned, relative to the structure center, back rotation of 90° brings the site 90/99 vectors closer to the vector site 92, but still 110° away from the primary vector (Fig. 3). Rotation 90° in the opposite sense would bring near coincidence with the primary vector, but there is no independent evidence to suggest that this is the true rotation style. If shock had in some way

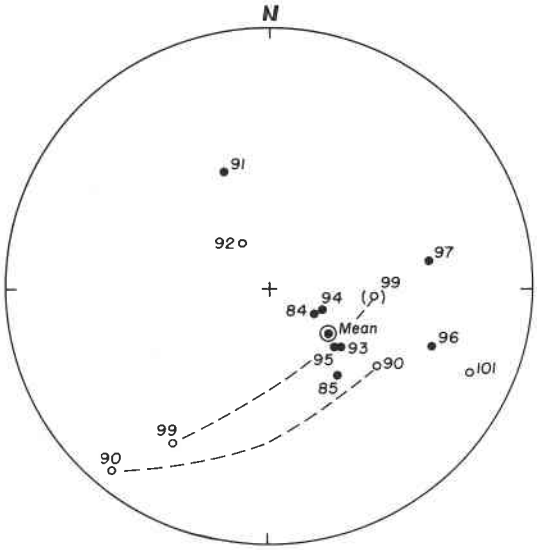


FIG. 3. Stereographic projection of remanent vectors at Charlevoix. Solid circles on lower hemisphere, open circles on upper hemisphere. Dotted line with arrow to circles enclosed in parentheses show back rotation of site 90 and 99 vectors, by 90° about horizontal NW axis. Double circle is mean of sites 84, 85, 93, 94, and 95.

caused self-reversal of the primary remanent vector, then associated deformation (uplift only at site 92) could explain these data. But although the intensity of most samples at sites 90 and 99 is significantly lower than that from all others, there is no obvious petrologic or mineralogic evidence of any change. In addition, this remanence is comparably stable to that from other sites, being virtually unaffected by AF demagnetization in fields up to 500 oe (Fig. 2). Shock is known to decrease the intensity of NRM (Hargraves and Perkins 1968), but without significantly affecting the orientation of the stable component, and certainly not causing reversal. Self-reversal plus rotation is thus an extremely ad hoc, if not inconceivable explanation, and the precise significance of these anomalous vectors remains unclear.

A virtual geomagnetic pole calculated from the impactite RM vector is generally intermediate between the lower Paleozoic poles and the present local virtual geomagnetic pole (Table 2). The significance of this remanent vector is thus doubtful, and it is of no aid in precise dating.

Conclusions

(i) The average remanent vector from five sites in anorthosite well removed (>20 km) from the Charlevoix cryptoexplosion structure, gives a VGP at 7° N, 32° W, δp 12°, δm 16°, similar to other Grenville poles.

(ii) The RM data indicates that rocks carrying this vector and located 15 to 20 km from the structure center have been rotated as a result of its formation; the vector from three sites within 15 km are anomalous, possibly reflecting modification (reversal?) of the primary vector, plus considerable rotation.

(iii) One site in 'impactite' gives generally inconclusive results.

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ELMORE, W. C. 1938. Ferromagnetic colloid for studying magnetic structures. Phys. Rev., **54**, pp. 309–310.

HARGRAVES, R. B. and PERKINS, W. E. 1968. Investigations of the effect of shock on natural remanent magnetism. J. Geophys. Res., **74**, pp. 2576–2589.

IRVING, E. and PARK, J. K. 1972. Hairpins and superintervals. Can. J. Earth Sci., **9**, pp. 1318–1324.

IRVING, E., PARK, J. K., and ROY, J. L. 1972. Paleomagnetism and the origin of the Grenville Front. Nature, **236**, pp. 344–346.

MAWDSLEY, J. B. 1927. St. Urbain area, Charlevoix district, Quebec. Geol. Surv. Can., Mem. 152, 58 p.

ROBERTSON, P. B. 1968. La Malbaie structure, Quebec, a Paleozoic meteorite impact site. Meteoritics, **4**, pp. 89–112.

RONDOT, J. 1966. Rapport preliminaire sur la region de la Malbaie, Charlevoix. Min. Rech. Nat. Que. R.P. No. 544, 19 p.

——— 1968. Nouuel impact meteoritique fossile? La structure semi-circulaire de Charlevoix. Can. J. Earth Sci., **5**, pp. 1305–1317.

——— 1971. Impactite of the Charlevoix structure, Quebec, Canada. J. Geophys. Res., **76**, pp. 5414–5423.

——— 1972. La transgression Ordovicienne dans le Comté de Charlevoix, Quebec. Can. J. Earth Sci., **9**, pp. 1187–1203.

ROSE, E. R. 1962. Le fer et le titane dans l'anorthosite de Saint Urbain, Quebec. Comm. Geol. Canada, Etude 61–7, 27 p.

——— 1972 Notes on remanent magnetism associated with some titaniferous deposits in Canada. Proc. Geol. Ass. Can., **24**, pp. 69–74.

STACEY, F. D. and BANERJEE, S. K. 1973. The physical principles of rock magnetism. Amsterdam, Elsevier, (In press).